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Prof. Jean Carlson and Prof. Holly Moeller
California NanoSystems Institute
University of California, Santa Barbara

Dear Professors Carlson and Moeller,

I am writing to express my strong interest in the Postdoctoral Researcher position in machine learning and computational modeling for the Wellcome Leap *Foundations of a Resilient Microbiome* project. Your plan to grow thousands of early-life gut communities in ExFAB, perturb them, and use models to move from correlation toward causation is exactly the kind of ambitious, collaborative science I want to be part of.

The first thousand days of life are one of the most critical windows for shaping long-term health, and my own research keeps bringing me back to that idea. During my PhD at UNAM's Institute of Biotechnology, I studied the gut microbiome of Mexican children with obesity and metabolic syndrome, which showed me how much the microbial communities of early life matter for later health. I would be excited to bring that perspective to neurodevelopment.

What defines my work most is its versatility. Across 15 peer-reviewed publications, I have worked with human, mouse, shrimp, bacterial, and plant systems, moving between different omics approaches as each question required. My dissertation followed gut communities from who is there to what they actually produce, identifying antimicrobial peptides encoded by plasmids and phages, while other projects ranged from the gut virome in childhood obesity to how environmental enrichment slowed dysbiosis in obese mice. Each new system taught me to adapt quickly, learn the language of new collaborators, and look for the principles that connect very different biological settings.

Most of my work has been computational, turning large and complex multi-omics datasets into biological insight, yet I am equally comfortable at the bench, and I enjoy working with experimental teams to turn their questions into testable models. I am eager to deepen my training in ecological theory, dynamical modeling, and causal inference, and I can think of no better place to do it than alongside your team.

I am based in San Diego, authorized to work in the U.S., and available to start on or after November 30, 2026. You can explore my portfolio and publications at www.lmgb.xyz. Thank you for your time, and I would love the opportunity to discuss how I can contribute to the *Foundations of a Resilient Microbiome* project.

Sincerely,